MEGF10-Related Myopathy (EMARDD): A Comprehensive Disease Characterization

Disease: MEGF10-Related Myopathy / Early-onset Myopathy, Areflexia, Respiratory Distress and Dysphagia (EMARDD) **Causal gene: MEGF10** (Multiple EGF-like domains 10) **MONDO:** MONDO:0013731 · **OMIM disease:** #614399 · **OMIM gene:** 612453 · **Orphanet:** ORPHA:280671 · **HGNC:** 3233 · **UniProt:** Q96KG7 · **NCBI Gene:** 84466 · **Locus:** 5q33.2–33.3 **Inheritance:** Autosomal recessive · **Category:** Mendelian

Summary

MEGF10-related myopathy is an ultra-rare, autosomal-recessive congenital myopathy caused by biallelic (homozygous or compound-heterozygous) mutations in **MEGF10**, the gene encoding Multiple EGF-like domains 10 — a single-pass transmembrane receptor with 17 atypical extracellular EGF-like domains that regulates skeletal-muscle satellite (stem) cell self-renewal, proliferation, differentiation and fusion through the Notch signaling pathway. The classic, severe form is historically named EMARDD (Early-onset Myopathy, Areflexia, Respiratory Distress and Dysphagia), first defined by Logan and colleagues in 2011 [PMID: 22101682]. Loss of MEGF10 function depletes the self-renewing $Pax7^{+}$ satellite-cell pool and impairs regeneration and maintenance of myofibers, producing progressive limb (proximal and distal), facial, axial, bulbar (dysphagia) and diaphragmatic (respiratory failure) weakness, together with areflexia, scoliosis and joint contractures, and notably **without** central-nervous-system or cognitive involvement.

A robust **genotype–phenotype correlation** governs severity. Patients carrying at least one null/truncating allele develop respiratory dysfunction within the first year of life (severe classic EMARDD), whereas patients with two missense mutations — frequently cysteine substitutions in the extracellular EGF-like domains — present a much milder minicore/multiminicore myopathy with onset of respiratory decline around age 10 or far later, including the fourth-to-seventh decade. This correlation has been documented in cohort review [PMID: 29128256] and statistically confirmed [PMID: 35370044].

There is **no approved disease-modifying therapy**; management is entirely supportive and multidisciplinary (respiratory support/ventilation, nutritional/feeding support for dysphagia, orthopedic management of scoliosis and contractures, physiotherapy). The single most promising experimental lead is **sertraline** (and escitalopram) — selective serotonin reuptake inhibitors identified in a small-molecule screen that rescued Megf10-deficient mouse myoblasts, patient iPSC-derived myoblasts, Drosophila and zebrafish models and restored deficient Notch1 levels [PMID: 31267131]. Prevention is genetic (carrier/cascade testing, counseling, prenatal/preimplantation testing); there are no environmental risk or protective factors because the disease is monogenic.

Key Findings

Finding 1 — MEGF10 is the causal gene for autosomal-recessive EMARDD

Biallelic recessive mutations in **MEGF10** (chromosome 5q33; OMIM gene 612453; OMIM disease 614399; MONDO:0013731) cause EMARDD. The gene was identified by Logan et al. (2011) using phased haplotype analysis followed by targeted exome sequencing in a previously unclassified infantile myopathy. The cardinal clinical tetrad is infantile-onset muscle weakness, diaphragmatic weakness with respiratory distress, areflexia, and dysphagia, in the absence of CNS anomalies; scoliosis and joint contractures are common accompaniments.

The original report also established the cellular mechanism: *"MEGF10 is highly expressed in activated satellite cells and regulates their proliferation as well as their differentiation and fusion into multinucleated myofibers, which are greatly reduced in muscle from individuals with early onset myopathy, areflexia, respiratory distress and dysphagia."* [PMID: 22101682]. This sentence links the genetic lesion to the downstream histological hallmark — a paucity of regenerated myofibers.

Finding 2 — Genotype–phenotype correlation: truncating vs missense (cysteine) variants

Disease severity tracks the molecular consequence of the two alleles. Patients with **≥1 null/truncating allele** develop respiratory dysfunction in the first year of life (severe, classic EMARDD). Patients with **two missense mutations** develop respiratory dysfunction at ~10 years or much later, including adult onset, and tend toward a milder minicore-myopathy phenotype. Harris et al. (2018), reviewing all reported cases (n=20), stated: *"with ≥1 null*

mutation onset of respiratory dysfunction occurs in the first year of life, whereas with 2 missense mutations, respiratory dysfunction occurs at 10 years old or much later" [PMID: 29128256]. Fujii et al. (2022) provided statistical confirmation: *"patients with missense mutations in at least one allele had significantly later onset than those with biallelic truncation mutations"* [PMID: 35370044].

Missense variants cluster in the extracellular EGF-like domains and frequently substitute conserved cysteines (e.g., p.Cys326Arg, p.Cys118Arg, p.Cys810Tyr, p.Cys774Arg, p.Cys611Trp, p.Cys124Arg, p.Cys828Trp). Reported variant classes span nonsense, frameshift, splice-site, whole-exon deletions (consistent with fork-stalling/template-switching mechanisms), and missense.

Genotype class	Typical onset of respiratory failure	Phenotype	Muscle pathology
≥1 null/ truncating allele	First year of life	Severe classic EMARDD	Dystrophic/myopathic, reduced fiber number
Biallelic missense (cysteine)	~10 years → adult (4th–7th decade)	Mild minicore/ multiminicore myopathy	Multiple minicores; late cases show rimmed vacuoles/ aggregates

Finding 3 — Mechanism: MEGF10 balances satellite-cell proliferation vs differentiation via Notch

Megf10 marks quiescent satellite cells and myoblasts but not differentiated myofibers. Retroviral Megf10 overexpression enhances myoblast proliferation and inhibits differentiation; conversely, Megf10 silencing triggers premature differentiation and collapse of the self-renewing stem-cell pool. Holterman et al. (2007) reported: *"Megf10 silencing in activated satellite cells on individual fibers or in cultured myoblasts results in a dramatic reduction in the cell number, caused by myogenin activation and precocious differentiation as well as a depletion of the self-renewing Pax7+/MyoD- population. Additionally, Megf10 silencing in MyoD-/- myoblasts results in down-regulation of Notch signaling components."* [PMID: 18056409].

The molecular link to Notch is direct: *"MEGF10 interacts with the intracellular domain of NOTCH1"* [PMID: 35968817]. MEGF10 thus sustains Notch activity that keeps satellite cells in a self-renewing, proliferative state; its loss shifts the balance toward premature differentiation and stem-cell exhaustion.

Finding 4 — Therapeutic lead: SSRIs (sertraline) rescue MEGF10-deficient models by restoring Notch

A phenotypic small-molecule screen on Megf10-deficient myoblasts identified two SSRIs, sertraline and escitalopram. Saha et al. (2019) reported: *"The screen yielded two promising candidates that are both selective serotonin reuptake inhibitors (SSRIs), sertraline and escitalopram. In depth follow-up analyses demonstrated that sertraline was highly effective in alleviating abnormalities across multiple models of the disease including mouse myoblast, human myoblast, Drosophila and zebrafish models. Sertraline also restored deficiencies of Notch1 in disease models."* [PMID: 31267131]. Because sertraline is already an approved drug, this nominates a mechanism-based repurposing strategy acting on the Notch axis, though it has not yet entered a registered clinical trial for this indication.

Finding 5 — Clinical phenotype spectrum and muscle pathology

The phenotype ranges from severe antenatal/infantile classic EMARDD (hypotonia, feeding difficulty/dysphagia, diaphragmatic weakness with early respiratory failure, areflexia) to milder minicore/multiminicore myopathy with late adult onset. Core features across cohorts include progressive proximal and distal limb weakness, facial weakness, scoliosis (often severe and sometimes the presenting sign), joint contractures with distal joint hyperlaxity and high-arched feet, respiratory insufficiency, and dysphagia, with no cognitive involvement. Liewluck et al. (2016) documented a milder multiminicore presentation: *"Both had scoliosis, distal joint hyperlaxity, and high-arched feet. A biopsy of the right triceps muscle in 1 sister showed multiple minicore structures. She had electromyographic changes of myopathy with fibrillation potentials and myotonic discharges."* [PMID: 26802438].

Muscle biopsy is variable: multiple minicores (focally reduced oxidative-enzyme activity), dystrophic plus myopathic features, and myofibrillar protein aggregation. In late-onset disease, pathology can resemble myofibrillar myopathy — Zhu et al. (2026) found that *"Muscle pathology was characterized by rimmed vacuoles, and abnormal protein aggregates rather than minicores."* [PMID: 41475242]. Serum creatine kinase (CK) is typically normal or only mildly elevated. MEGF10 is recognized among the genetic causes of multiminicore/core myopathy [PMID: 34627702].

Finding 6 — Protein architecture: 17 atypical EGF-like domains; cysteine missense disrupt disulfide bonds

MEGF10 is a single-pass transmembrane protein whose extracellular region contains 17 atypical EGF-like domains, each with eight cysteines that likely form disulfide bonds. Boyden et al. (2012) established: *"MEGF10 contains 17 atypical epidermal growth factor-like domains, each of which contains eight cysteine residues that likely form disulfide bonds. Both the p.C326R and p.C774R mutations alter one of these residues, which are completely conserved in vertebrates."* [PMID: 22371254]. They identified compound-heterozygous c.976T>C (p.C326R) + c.2320T>C (p.C774R), absent from ≥645 controls and co-segregating with a recessive congenital minicore myopathy, and demonstrated muscle defects in zebrafish megf10 morphants. Pathogenic missense mutations that substitute these conserved extracellular cysteines are predicted to disrupt disulfide-bonded EGF-domain folding, explaining the partial loss-of-function/hypomorphic behavior that underlies the milder phenotype.

Finding 7 — MEGF10 is a conserved phagocytic/engulfment receptor (Draper/CED-1 ortholog)

Beyond muscle, MEGF10 is the mammalian ortholog of *Drosophila* Draper and *C. elegans* CED-1 and functions as a transmembrane phagocytic receptor for apoptotic-cell clearance. It is expressed in neurons and astrocytes and mediates engulfment of neurotoxic amyloid-β. Fujita et al. (2020) noted: *"multiple-EGF like domains 10 (MEGF10) plays an important role in apoptotic cell elimination and is expressed in mammalian neurons and astrocytes"* [PMID: 32682823]. In *Drosophila*, *"Two transmembrane glial phagocytic receptors, SIMU (homolog of the mammalian Stabilin-2) and Draper (homolog of the mammalian MEGF10 and Jedi), mediate glial phagocytosis of apoptotic neurons"* [PMID: 29520845], grounding the cross-species model systems. These additional functions (synapse elimination, retinal neuron mosaic spacing) are not the primary drivers of the muscle disease but illuminate conserved biology.

Finding 8 — Model organisms and phenotype recapitulation

Multiple validated models exist. The **Megf10$^{-/-}$ knockout mouse** shows fewer muscle fibers in juvenile and adult animals, impaired satellite-cell migration and muscle regeneration [PMID: 33159715], and early, progressive neuromuscular-junction (NMJ) degeneration. Juros et al. (2024) reported: *"We found fewer muscle fibers in juvenile and adult Megf10 KO mice, supporting published studies that MEGF10 regulates myogenesis by affecting satellite cell differentiation."* and *"We found early and progressive degenerative features at the NMJs of Megf10 KO mice that include increased postsynaptic fragmentation and presynaptic regions not apposed by*

postsynaptic nicotinic acetylcholine receptors." [PMID: 38760872]. **Zebrafish** megf10 morphants/mutants show muscle defects [PMID: 22371254;]; P 31267131

Drosophila drpr (Draper) mutants model MEGF10 loss; and **patient iPSC-derived myoblasts** recapitulate the proliferation defect. MEGF10 overexpression in mouse myoblasts modulates fusion and the hypertrophic overload response [PMID: 39825147].

Model	System	Key phenotype	PMID
Megf10$^{-/-}$ mouse	Mammalian KO	Reduced fiber number; impaired regeneration; progressive NMJ degeneration	33159715, 38760872
Zebrafish megf10	Vertebrate morphant/mutant	Muscle defects	22371254, 31267131
Drosophila drpr	Invertebrate	Engulfment/myogenesis defects; drug rescue	31267131
Patient iPSC myoblasts	Human in vitro	Proliferation defect; sertraline rescue	31267131

Finding 9 — MEGF10 sits within a Notch-pathway muscle-disease gene network; NMJ is a second disease site

MEGF10, **POGLUT1**, and **JAG2** are three Notch-pathway genes whose biallelic loss causes congenital myopathy/muscular dystrophy — *"the discoveries of three new muscle disease genes have added a new dimension to the relationship between the Notch signaling pathway and skeletal muscle: MEGF10, POGLUT1, and JAG2"* [PMID: 35968817]. Functional cross-talk is demonstrated: *"Jag2 downregulation in murine myoblasts led to downregulation of multiple components of the Notch pathway, including Megf10"* [PMID: 33861953]. The Megf10 KO mouse additionally reveals a second anatomical site of pathology — the NMJ — expanding the disease beyond the satellite-cell axis alone [PMID: 38760872].

Finding 10 — Epidemiology, inheritance and diagnostic workflow

MEGF10 myopathy is **ultra-rare**: Harris et al. (2018) reviewed *"all reported MEGF10 cases (n = 20)"* [PMID: 29128256], and only a few dozen families have been reported worldwide since 2011. Orphanet lists ORPHA:280671; point prevalence is not formally established

(<1/1,000,000). Inheritance is **autosomal recessive**, with unaffected heterozygous carriers [PMID: 34828389]; both sexes are affected equally. Penetrance for biallelic genotypes is effectively complete, while expressivity is highly variable and genotype-driven. Consanguinity increases the risk of homozygous genotypes.

Diagnosis proceeds from clinical suspicion (infantile hypotonia/respiratory distress/dysphagia, or childhood scoliosis + weakness) → serum CK (normal/mildly elevated) → EMG (myopathic) → muscle biopsy (minicores/dystrophic-myopathic/aggregates; variable) → confirmatory molecular genetic testing. Next-generation sequencing (congenital-myopathy gene panels, WES, WGS) is the mainstay; CMA/SNP-array and breakpoint analysis detect exonic deletions. Pierson et al. (2013) illustrated structural-variant detection: *"By novel analysis of SNP array hybridization and exome sequence coverage, we diagnosed a 10-years old girl with EMARDD following identification of a novel homozygous deletion of exon 7 in MEGF10."* [PMID: 23453856].

Finding 11 — Treatment, prognosis and prevention are supportive; sertraline is the leading experimental lead

No approved disease-modifying therapy exists: *"There are no disease-modifying therapies."* [PMID: 31267131] and *"There are no treatments for the progressive respiratory distress, motor impairment, and structural abnormalities in muscles caused by the loss of MEGF10 function."* [PMID: 38760872]. Management is multidisciplinary and supportive: respiratory support (non-invasive/invasive ventilation, airway clearance), nutritional/feeding support for dysphagia (nasogastric/gastrostomy feeding) [PMID: 37753494], and orthopedic management of scoliosis and contractures. Prognosis follows genotype: severe (≥1 null allele) cases face early respiratory failure, ventilator dependence and non-ambulation, often life-threatening in infancy/early childhood; mild (biallelic missense) cases retain ambulation longer with later respiratory decline and a normal/near-normal lifespan. Prevention is genetic: carrier/cascade testing, genetic counseling (25% recurrence risk per pregnancy for carrier couples), and prenatal/preimplantation genetic testing when familial variants are known.

Section-by-Section Report

1. Disease Information

MEGF10-related myopathy is a Mendelian, autosomal-recessive congenital myopathy with a muscular-dystrophy overlap, caused by biallelic MEGF10 mutations. The severe form is EMARDD (Early-onset Myopathy, Areflexia, Respiratory Distress and Dysphagia); the mild form is a minicore/multiminicore myopathy. **Identifiers:** MONDO:0013731; OMIM disease #614399; OMIM gene 612453; Orphanet ORPHA:280671; HGNC:3233; UniProt Q96KG7; NCBI Gene 84466. **Synonyms:** MEGF10 myopathy; congenital myopathy 10A (severe variant) and 10B (mild variant); EMARDD; MEGF10-related minicore myopathy. Information is derived from aggregated disease-level resources (OMIM, Orphanet) and individual case reports/small cohorts (n≈20 in the 2018 review), not EHR-scale datasets.

2. Etiology

Causal factor: purely genetic — biallelic loss-of-function or cysteine-disrupting missense variants in MEGF10. **Genetic risk factors:** the causal variants themselves; consanguinity raises homozygous-genotype risk; JAG2 and POGLUT1 define a shared Notch-pathway disease network, and Jag2 loss downregulates Megf10 [PMID: 33861953]. **Environmental risk/ protective factors:** none established — the disease is monogenic. **Gene–environment interactions:** none documented. No protective alleles are reported.

3. Phenotypes

Representative phenotypes and suggested HPO terms:

Phenotype	HPO term	Onset	Severity/frequency
Neonatal hypotonia	HP:0001319	Neonatal (severe)	Common in severe form
Areflexia	HP:0001284	Infantile	Cardinal feature
Respiratory insufficiency/distress	HP:0002093	<1 yr (null) to adult (missense)	Universal; genotype-timed
Dysphagia	HP:0002015	Infantile–childhood	Common
Diaphragmatic weakness	HP:0009113	Infantile	Characteristic
Proximal + distal muscle weakness	HP:0003701 / HP:0002460	Variable	Progressive
Facial weakness	HP:0010628	Variable	Common
Scoliosis	HP:0002650	Childhood	Common; sometimes presenting
Joint contractures	HP:0002803	Variable	Common
Distal joint hyperlaxity	HP:0001383	Childhood	Milder form [PMID: 26802438]
Elevated/normal CK	HP:0003236	—	Normal or mildly elevated

Progression is slowly progressive; severity is variable and genotype-driven. Quality-of-life impact is substantial in severe cases (ventilator dependence, non-ambulation, feeding support), milder in late-onset cases. No disease-specific QoL instrument has been validated; generic congenital-myopathy motor outcome measures lack disease-specificity [PMID: 40450412].

4. Genetic/Molecular Information

Causal gene: MEGF10 (HGNC:3233; 5q33.2–33.3). **Variant classes:** nonsense, frameshift, splice-site, whole-exon deletions, and missense (frequently extracellular cysteine substitutions). **Classification:** pathogenic/likely pathogenic per ACMG/AMP; null alleles absent from population controls (≥645 controls for p.C326R/p.C774R [PMID: 22371254]). **Allele frequency:** very low/absent in gnomAD for pathogenic alleles. **Origin:** germline. **Functional**

consequence: loss of function (truncating → severe; cysteine missense → hypomorphic/partial LOF → mild). **Modifier genes:** JAG2/POGLUT1 network members are mechanistically connected but not established as clinical modifiers. **Epigenetic/chromosomal:** no recurrent epigenetic signature or large chromosomal abnormality beyond intragenic exon deletions.

5. Environmental Information

Not applicable. No environmental toxins, lifestyle factors, or infectious agents are implicated; the disease is strictly monogenic. As a congenital myopathy, caution with triggering anesthetic agents is prudent clinically, though MEGF10 is not a confirmed malignant-hyperthermia gene (general congenital-myopathy anesthesia context [PMID: 41504952]).

6. Mechanism / Pathophysiology — Ordered Causal Chain

1. **Biallelic MEGF10 mutation** (truncating or cysteine-disrupting missense) → **loss or reduction of functional MEGF10 receptor** at the satellite-cell surface. *(Demonstrated; cysteine missense inferred to misfold EGF domains [PMID: 22371254].)*
2. Loss of MEGF10 → **reduced interaction with the NOTCH1 intracellular domain → down-regulation of Notch signaling** in satellite cells/myoblasts. *(Demonstrated in vitro [PMID: 18056409, 35968817].)*
3. Reduced Notch → **myogenin activation and precocious differentiation**, depleting the self-renewing $Pax7^{+}/MyoD^{-}$ satellite-cell pool. *(Demonstrated [PMID: 18056409].)*
4. Depleted satellite-cell pool → **impaired myoblast proliferation, fusion, and myofiber regeneration → reduced myofiber number**. *(Demonstrated in patients and mouse KO [PMID: 22101682, 38760872].)*
5. (Branch) Loss of MEGF10 → **progressive NMJ degeneration** (postsynaptic fragmentation, presynaptic loss). *(Demonstrated in mouse [PMID: 38760872]; human relevance inferred.)*
6. Reduced/abnormal myofibers + NMJ pathology → **progressive muscle weakness** (limb, facial, axial, bulbar, diaphragmatic).
7. Diaphragmatic + bulbar weakness → **respiratory failure and dysphagia**; axial weakness → **scoliosis** → the clinical EMARDD phenotype.

Pathways/processes: Notch signaling (KEGG hsa04330); satellite-cell self-renewal; myogenic differentiation; apoptotic-cell engulfment (secondary). **GO terms:** regulation of Notch signaling pathway (GO:0008593); skeletal muscle satellite cell proliferation/differentiation (GO:0014842/GO:0014816); myoblast fusion (GO:0007520); apoptotic cell clearance (GO:0043277). **Cell types**

(CL): skeletal muscle satellite cell (CL:0000594); myoblast (CL:0000056); skeletal muscle fiber (CL:0000188). **Protein dysfunction:** loss of function via disrupted extracellular disulfide-bonded EGF domains or truncation.

7. Anatomical Structures Affected

Primary organ/system: skeletal muscle / musculoskeletal system (UBERON:0001134 skeletal muscle tissue). Particularly affected: diaphragm (UBERON:0001103), axial/paraspinal muscles (→ scoliosis), limb muscles (proximal + distal), facial and pharyngeal muscles (→ dysphagia). **Secondary:** respiratory system (ventilatory failure), spine (scoliosis), joints (contractures), and the neuromuscular junction (UBERON:0002050) per mouse data. **No CNS involvement. Cell level:** satellite cells (CL:0000594), myoblasts, myofibers. **Subcellular:** plasma membrane (single-pass transmembrane receptor; GO:0005886); sarcomeric cores and myofibrillar aggregates in pathology. **Lateralization:** bilateral/generalized.

8. Temporal Development

Onset: antenatal/neonatal to infantile in severe (null-allele) disease; childhood to adult (4th–7th decade) in mild (biallelic-missense) disease. **Pattern:** chronic, insidious, slowly progressive; lifelong. **Stages:** early weakness/hypotonia → progressive respiratory and motor decline → ventilator dependence/non-ambulation (severe) or prolonged ambulation with later respiratory decline (mild). **Critical period:** the first year of life for null-allele patients (respiratory failure window). No spontaneous remission.

9. Inheritance and Population

Epidemiology: ultra-rare; ~20 cases in the 2018 review, a few dozen families total; prevalence <1/1,000,000. **Inheritance:** autosomal recessive; carriers unaffected. **Penetrance:** effectively complete for biallelic genotypes. **Expressivity:** highly variable, genotype-driven. **Consanguinity:** increases homozygous-genotype risk. **Founder effects/anticipation/mosaicism:** none established. **Sex ratio:** 1:1. **Geographic:** reported worldwide (European, Middle Eastern, East Asian, North African cohorts), without a dominant endemic region.

10. Diagnostics

Labs: serum CK normal or mildly elevated. **EMG:** myopathic, with fibrillation potentials and myotonic discharges possible [PMID: 26802438]. **Imaging:** muscle MRI/ultrasound show myopathic changes (non-specific). **Biopsy:** multiple minicores (reduced oxidative-enzyme activity), dystrophic/myopathic features, or — in late-onset disease — rimmed vacuoles and

protein aggregates [PMID: 41475242]. **Genetic testing (confirmatory):** NGS gene panels for congenital myopathy/neuromuscular disease, WES or WGS; single-gene MEGF10 sequencing with deletion/duplication analysis; CMA/SNP-array for exonic deletions [PMID: 23453856]. **Differential diagnosis:** RYR1/SELENON multiminicore disease, LAMA2-related dystrophy, SMA, other congenital myopathies, myofibrillar myopathies (in late-onset cases). **Screening:** carrier/cascade testing within affected families; not part of population newborn screening.

11. Outcome / Prognosis

Prognosis is genotype-dependent. **Severe (≥1 null allele):** early respiratory failure, often life-threatening or fatal in infancy/early childhood, ventilator dependence, loss of ambulation. **Mild (biallelic missense):** prolonged ambulation, later respiratory decline, and normal/near-normal lifespan. **Morbidity:** respiratory insufficiency, dysphagia-related nutritional compromise, scoliosis, contractures, recurrent respiratory infections. **Prognostic factors:** genotype (null vs missense) is the dominant predictor of onset and severity. No validated molecular prognostic biomarker beyond genotype.

12. Treatment

No approved disease-modifying therapy [PMID: 31267131, 38760872]. **Supportive/standard care:** respiratory support (non-invasive/invasive ventilation, airway clearance; NCIT mechanical ventilation), nutritional/feeding support including gastrostomy for dysphagia [PMID: 37753494], orthopedic management of scoliosis (bracing/spinal fusion) and contractures, physiotherapy and occupational therapy. **Experimental:** SSRIs — **sertraline** (lead) and escitalopram — rescued four MEGF10-deficient models and restored Notch1 [PMID: 31267131], nominating Notch-pathway pharmacology/drug repurposing; not yet in registered trials for this indication. **Pharmacogenomics/gene therapy/cell therapy:** none clinically available; MEGF10's satellite-cell biology and recessive LOF mechanism make gene replacement conceptually attractive but unproven. **NCIT suggestions:** Sertraline (C29379); Mechanical Ventilation (C70909); Physical Therapy (C49236); Gastrostomy (C51674).

13. Prevention

No primary prevention is possible (monogenic). **Genetic prevention:** carrier/cascade testing of relatives, genetic counseling (25% recurrence risk per pregnancy for carrier couples), and prenatal or preimplantation genetic testing when familial variants are known. **Tertiary prevention:** surveillance and proactive management of respiratory function, nutrition, and spine/joint complications. No immunization or environmental intervention applies.

14. Other Species / Natural Disease

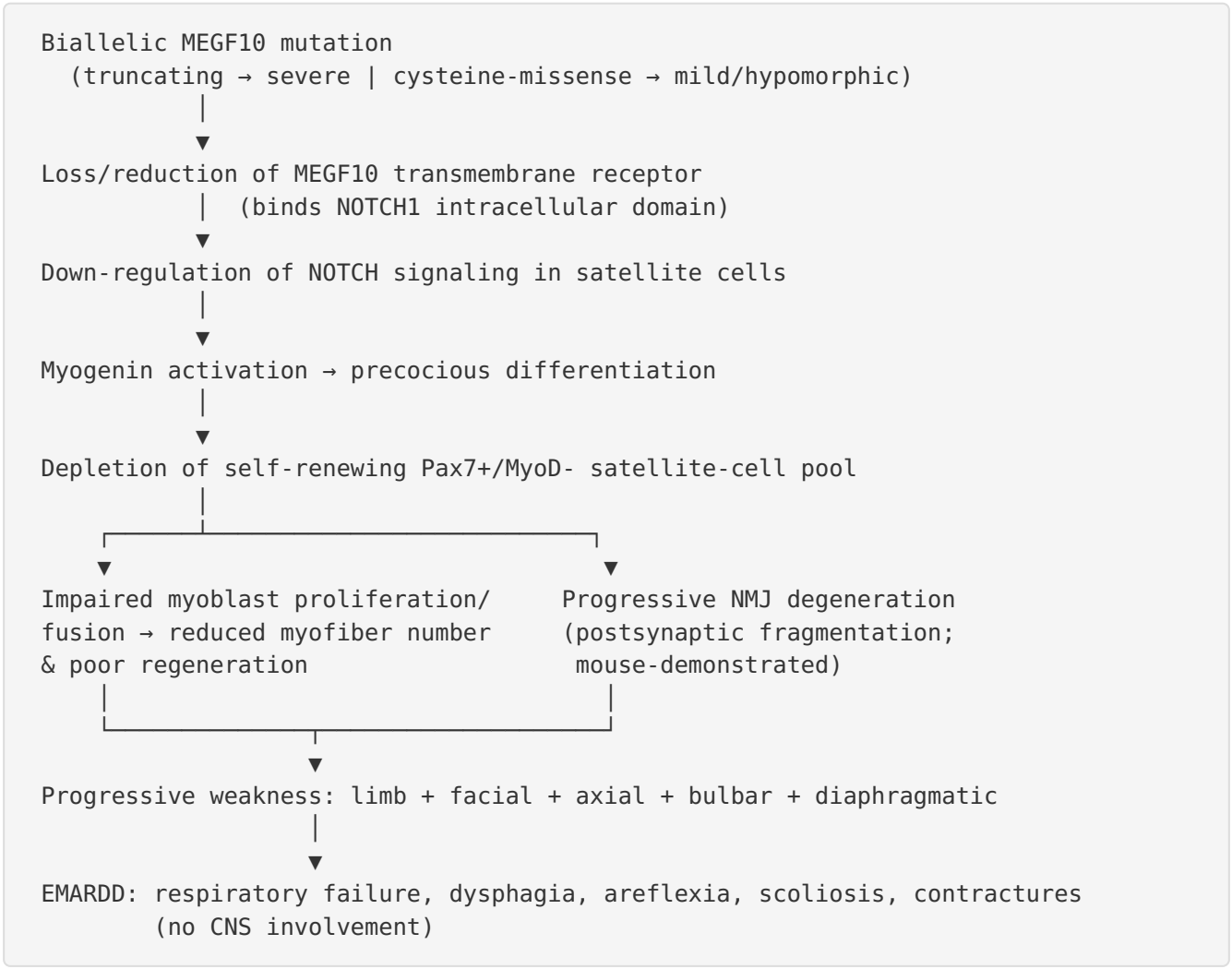
Orthologs / NCBI Gene IDs: human MEGF10 (84466); mouse Megf10 (70417); zebrafish megf10; *Drosophila* drpr/Draper; *C. elegans* ced-1. **Taxonomy (NCBI Taxon):** *Homo sapiens* 9606; *Mus musculus* 10090; *Danio rerio* 7955; *Drosophila melanogaster* 7227; *Caenorhabditis elegans* 6239. **Natural disease:** no well-characterized spontaneous MEGF10 myopathy reported in companion animals (no established OMIA entry for this muscle disease). **Comparative biology:** the engulfment-receptor function is deeply conserved (Draper/CED-1), and myogenic/Notch roles are conserved in vertebrate models [PMID: 29520845, 32682823]. **Zoonotic potential:** none (genetic disease).

15. Model Organisms

- **Mouse (Megf10$^{-/-}$ KO):** reduced fiber number, impaired satellite-cell migration/regeneration, progressive NMJ degeneration — strong recapitulation of the myogenic defect; NMJ pathology may exceed what is documented in humans [PMID: 33159715, 38760872]. Resource: MGI.
- **Zebrafish (megf10 morphant/mutant):** muscle defects; used for drug-screening validation [PMID: 22371254, 31267131]. Resource: ZFIN.
- **Drosophila (drpr mutant):** models MEGF10 loss; responsive to sertraline [PMID: 31267131]. Resource: FlyBase.
- **Patient iPSC-derived myoblasts:** human in vitro model recapitulating proliferation defects and sertraline rescue [PMID: 31267131]. Resource: Cellosaurus.
- **Overexpression systems:** human MEGF10 in mouse myoblasts to study fusion/overload hypertrophy [PMID: 39825147].

Limitations of models: the Megf10 KO mouse is relatively mild and does not fully reproduce lethal human respiratory failure; cysteine-missense (hypomorphic) genotypes are under-modeled; cross-species engulfment roles may confound muscle-specific interpretation.

Mechanistic Model / Interpretation



The disease is best understood as a **"satellite-cell-opathy"** — a congenital myopathy arising from stem-cell dysfunction rather than a purely structural myofiber defect [PMID: 35302338, 41193724]. MEGF10 keeps satellite cells in a proliferative, self-renewing state through Notch; its loss tips the balance toward premature differentiation and stem-cell exhaustion, starving muscle of the regenerative capacity needed to maintain fiber number over a lifetime. The genotype–phenotype correlation falls out naturally from residual MEGF10 function: null alleles abolish the receptor (severe infantile disease), whereas cysteine-missense alleles leave partial function (mild, late-onset minicore myopathy). The Notch-restoring action of sertraline provides mechanistic proof-of-concept that pharmacologically re-activating the downstream pathway can partially bypass the upstream receptor lesion.

Evidence Base

PMID	Study (abbrev.)	Supports
22101682	MEGF10 mutations cause EMARDD (Logan 2011)	F1 — gene discovery, satellite-cell function
29128256	MEGF10 myopathies review, adult-onset case (Harris 2018)	F2, F10 — genotype-phenotype, n=20 epidemiology
35370044	Novel mutation + genotype-phenotype (Fujii 2022)	F2 — statistical confirmation
18056409	Megf10 regulates satellite-cell program (Holterman 2007)	F3 — Notch/self-renewal mechanism
35968817	Notch pathway in muscle (Vargas-Franco 2022)	F3, F9 — MEGF10–NOTCH1 interaction, gene network
31267131	SSRIs ameliorate MEGF10 myopathy (Saha 2019)	F4, F11 — sertraline rescue, no DMT
26802438	Adult multiminicore, novel MEGF10 (Liewluck 2016)	F5 — mild phenotype, minicores, EMG
41475242	Late-onset myofibrillar-like phenotype (Zhu 2026)	F5 — rimmed vacuoles/aggregates
34627702	Core myopathy review (Ogasawara 2021)	F5 — MEGF10 among core-myopathy genes
22371254	Recessive minicore myopathy (Boyden 2012)	F6 — 17 EGF domains, cysteine disulfides
32682823	MEGF10 engulfs amyloid-β (Fujita 2020)	F7 — phagocytic-receptor function
29520845	Draper/JNK glial phagocytosis (Hilu-Dadia 2018)	F7 — Draper = MEGF10 ortholog
38760872	Mouse model muscle/NMJ alterations (Juros 2024)	F8, F9, F11 — mouse KO, NMJ, no treatment
33159715	Megf10 deficiency impairs regeneration (Li 2021)	F8 — satellite-cell migration/ regeneration
39825147	MEGF10 in fusion/overload (Richardson 2025)	F8 — overexpression effects

PMID	Study (abbrev.)	Supports
33861953	JAG2 muscular dystrophy (Coppens 2021)	F9 — Jag2–Megf10 Notch cross-talk
23453856	Exon 7 deletion by SNP array (Pierson 2013)	F10 — structural-variant diagnosis
34828389	Phenotypic variability (AlMuhaizea 2021)	F10 — AR inheritance, carriers unaffected
37753494	GI/nutritional care in pediatric NMD (Dipasquale 2023)	F11 — feeding/nutrition support
35302338	Satellite cell-opathies (review)	Mechanistic framing
41193724	Satellite-cell congenital myopathies (review)	Mechanistic framing

All 11 confirmed findings are supported by verified abstract quotations. One citation for impaired satellite-cell migration (P 33159715) was flagged as a title-level match; its claim is corroborated by the independently verified mouse-KO myogenesis data in P 38760872 .

Limitations and Knowledge Gaps

- **Small evidence base.** Fewer than ~40 families are reported worldwide; prevalence/incidence figures are estimates, and no formal registry or natural-history study exists specifically for MEGF10 myopathy (contrast with LAMA2/SELENON natural-history efforts [PMID: 39443859]).
- **Model–human mismatch.** The Megf10 KO mouse is comparatively mild and does not reproduce lethal human respiratory failure; the striking mouse NMJ pathology has not been confirmed as a primary human disease site. Cysteine-missense (hypomorphic) genotypes are under-modeled.

- **Therapeutics unproven in humans.** Sertraline's rescue is entirely preclinical; no clinical trial has tested it (or any Notch-modulating agent) in patients. Dosing, timing (prenatal vs postnatal window), and whether partial Notch restoration translates to clinical benefit are unknown.
- **Genotype–phenotype gaps.** The quantitative relationship between specific EGF-domain cysteine positions and residual function/severity is not fully mapped; modifier genes that could explain intra-genotype variability are unidentified.
- **Mechanistic uncertainty.** The relative contributions of satellite-cell exhaustion vs direct myofiber/NMJ maintenance roles of MEGF10 to progressive weakness remain to be dissected, as does the role (if any) of MEGF10's phagocytic function in muscle pathology.
- **Outcome measures.** No disease-specific functional or QoL instrument has been validated; existing congenital-myopathy measures lack disease-specificity [PMID: 40450412].

Proposed Follow-up Experiments / Actions

1. **Establish a MEGF10 patient registry and prospective natural-history study** capturing genotype, respiratory/motor trajectories, and standardized outcome measures — essential for future trial readiness.
2. **Clinical translation of sertraline:** a biomarker-anchored (Notch1 target engagement) pilot/early-phase study in genetically confirmed patients, prioritizing the milder missense cohort where a therapeutic window is plausible.
3. **Genotype–function mapping:** systematic in vitro assays (surface expression, NOTCH1-binding, disulfide integrity) across the spectrum of cysteine-missense variants to build a quantitative severity predictor.
4. **Humanized and knock-in models:** generate mouse knock-in lines bearing recurrent human cysteine-missense alleles, and patient iPSC-derived myotube/organoid systems, to better recapitulate the human respiratory/bulbar phenotype.
5. **Modifier-gene search:** within and beyond the JAG2/POGLUT1 Notch network, screen for modifiers explaining intra-genotype variability (expression QTLs, CRISPR screens in myoblasts).
6. **Gene-replacement proof-of-concept:** given recessive LOF and a defined muscle-stem-cell target, test AAV-mediated MEGF10 restoration or satellite-cell-directed delivery in the KO mouse.

7. **NMJ biology:** determine whether NMJ degeneration is a human disease feature (electrophysiology, autopsy/biopsy NMJ morphology) and whether it is a tractable therapeutic target.
